

WASP-36b

R-0816

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_231226 · created 2026-07-28T13:11:13+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460304.9332 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.036
Transit depth (Rp/Rs)^2	2.77 +/- 1.19 [%]
Orbital Inclination (inc)	81.71 +/- 1.96
Airmass coefficient 1 (a1)	1.055 +/- 0.017
Airmass coefficient 2 (a2)	-0.036 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#1 - [560, 63]
Optimal Method	PSF photometry
Transit Duration (day)	0.062 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.167 $\pm$ 0.036 vs 0.1368 (deviation 0.67 σ)
Duration fit vs published	0.062 d vs 0.0758 d (deviation 0.48 σ)
Mid-time O–C	13.3 min
Depth SNR	3.7
Residual scatter	1.64 %

Light curve

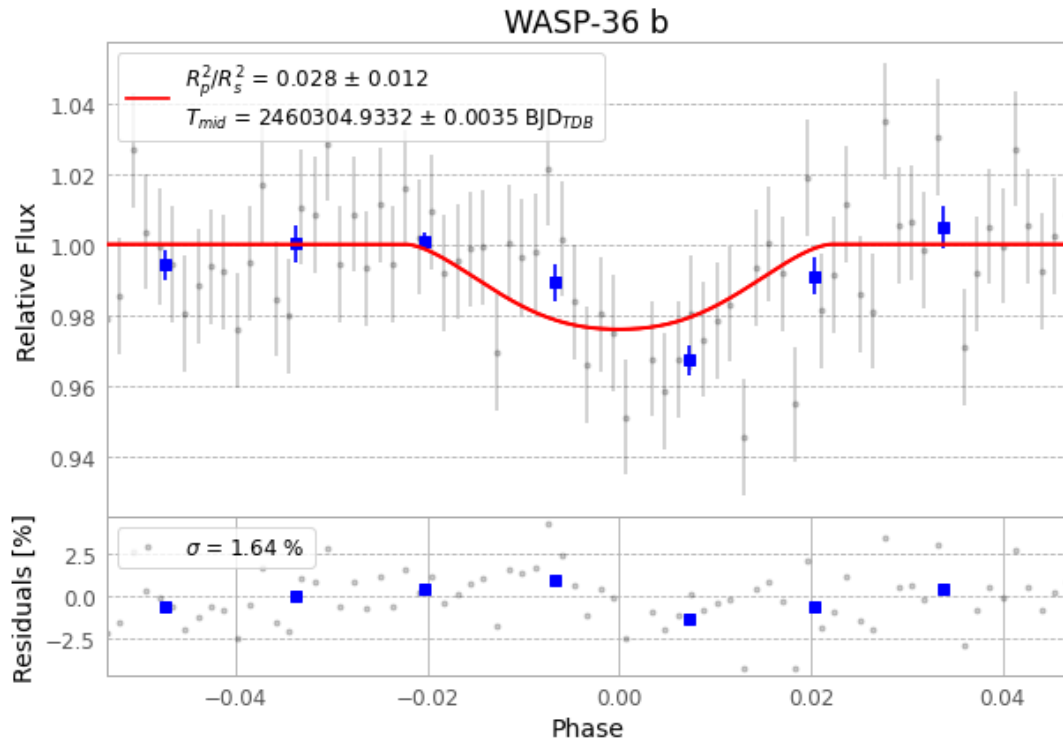


Figure 1 — FinalLightCurve_WASP-36 b_2023-12-26.png (SHA-256 519037b0ade6fb7f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17d091842754f78d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), WASP-36231226081815.FITS → WASP-36231226120015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-231226.log (83fd2bef0b86...), FinalLightCurve_WASP-36 b_2023-12-26.png (519037b0ade6...), FinalLightCurve_WASP-36 b_2023-12-26.csv (ecde2b374eeb...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 13:11Z.